Fundamentals of Natural Language Processing: Project Report

Automated ICD-10 Codification of Clinical Text

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Abstract

This report details the development of an automated Natural Language Processing (NLP) system designed to assign International Classification of Diseases (ICD-10) alphanumeric categories to short, unstructured Spanish medical literals. The proposed architecture employs a rigorous rule-based normalization pipeline to mitigate noise and clinical shorthand, followed by a prioritized inference strategy that combines direct regular expression matching, a flat Linear Support Vector Classifier (LinearSVC) built on TF-IDF features, and a cosine similarity fallback mechanism.

1 Introduction

The International Classification of Diseases provides a globally standardized taxonomy for clinical diagnoses (such as diseases and procedures), in accordance with the World Health Organization (WHO). Each condition is assigned a structured alphanumeric code organized in a hierarchical architecture of chapters, sections, and categories. The underlying rationale of employing this coding system is based on converting medical terminology (i.e., literals) into machine-readable structured form, which would allow analyzing disease incidence, making Electronic Medical Records (EMR) interoperable, and implementing Diagnosis Related Groups reimbursement system. The consequences of mistakes in applying ICD codes include improper distribution of healthcare budgets and erroneous epidemiological statistics.

Traditionally, assignment of ICD codes was done by trained clinical coders manually, which takes a considerable amount of time, entails inaccuracies, and demands regular re-training due to continuous development of the taxonomy, as there are over 142 thousand codes used in the newest version, ICD-10-CM [1]. This is why there have been extensive studies on automating this process from applying rules and machine learning algorithms (like SVM with TF-IDF vectors) to deep learning approaches, such as CNN with label-wise attention, GNN taking advantage of ICD hierarchy structure, and pretrained language models like BioBERT or ClinicalBERT [1].

The objective of this project is to accurately predict the first character of the corresponding CIE-10-ES (the Spanish equivalent of ICD-10) code given a brief Spanish medical literal (e.g., mapping the input *miocardiopatía dilatada* to the category I).

2 Methodology

2.1 Data Sources and Aggregation

To establish a robust feature space across 36 target classes, the training dataset was synthesized from three primary sources, yielding a combined training matrix of 201,809 samples:

- **Training Codification Data (`codification_data.csv`):** 13,700 explicitly labeled clinical pairs.
- **ICD Knowledge Base (`icd_d_p_pairs.csv`):** 179,742 official ICD-10 description-to-code mappings, providing standardized medical vocabulary.
- **CodiEsp Dataset:** An external corpus contributing 8,367 real-world Spanish clinical mentions to improve generalization. This public database was included to try to improve the model, and it is not mandatory to use it.

The main training and testing datasets were provided by *ASHO AI* and they represent real data. The primary challenges inherent to this dataset include:

- **Noisy, Informal Language:** Clinicians frequently utilize informal syntax under time constraints.
- **Bilingual Vocabulary:** The corpus contains a heterogeneous mix of Spanish (Castellano) and Catalan terminology.
- **High Density of Abbreviations:** The heavy reliance on non-standard medical shorthand (e.g., *HTA* for *hipertension arterial*) introduces significant issues, as the model cannot properly learn the relation between the abbreviation and other related diseases or procedures that are normally written in full form.
- **Contextual Brevity:** Inputs typically range from only 1 to 4 tokens, providing minimal context for machine learning models.

2.2 Preprocessing Pipeline

Given the extreme brevity of the text, predictive accuracy relies heavily on lexical normalization. The pipeline executes four sequential transformations prior to vectorization:

1. **Regex ICD Extraction:** Heuristics scan for explicit code mentions within the text. ICD-9 numeric codes are mapped to the nearest ICD-10 chapter letter via a localized lookup table.
2. **Normalization:** Text is lowercased and unicode accents are stripped (e.g., *prótesis mama* normalizes to *protesis mama*) to reduce feature sparsity.

3. **Cross-Lingual Translation:** A bespoke dictionary intercepts and translates frequent Catalan terms into standard Castellano (e.g., *cos estrany* → *cuerpo extrano*).

4. **Abbreviation Expansion:** Lexical analysis identified 564 candidate abbreviations, out of which 206 were manually extracted for replacement using an online dictionary of medical terms. Leveraging the SEDOM [2] medical dictionary, high-frequency tokens such as *HTA* (82 occurrences) and *IAM* are deterministically expanded into their full semantic forms (*hipertension arterial* and *infarto agudo miocardio*).

Table 1 presents a few literals before and after the preprocessing; it clearly illustrates the transitioning from the literals with accents, case-sensitive, loaded with abbreviation, to the cleaner version ready to be further processed by the model.

Table 1: Preprocessing Transformation Examples

Raw Literal	**Processed Output**
Hiperreactividad bronquial	hiperreactividad bronquial
miocardiopatía dilatada	miocardiopatia dilatada
HTA irc 6	hipertension arterial insuficiencia renal cronica 6
Crisis febriles atípicas	crisis febriles atipicas

2.3 Classification Architecture

As presented during the initial follow-up, the first iteration of the project was supposed to use the TF-IDF features alongside a Logistic Regression model (`LogisticRegression`). However, this approach was refined in an iteration of the solution and was replaced by a better strategy. Now, the core learning algorithm applies a flat, multi-class Linear Support Vector Classifier (`LinearSVC`).

2.3.1 Feature Engineering

Text representations are constructed using a `FeatureUnion` of two distinct `TfidfVectorizer` instances optimized with sublinear term frequency scaling:

- **Word N-grams (1-3):** Capped at 20,000 maximum features to capture standard medical terminology and short clinical phrases.
- **Character N-grams (2-6):** Utilizing a word-boundary analyzer (`char_wb`) capped at 40,000 features. This captures root morphologies and suffix variations (e.g., *-itis*) independently of stemming.

2.3.2 Prioritized Inference Strategy

To maximize accuracy, prediction during inference follows a strict three-tier priority logic:

1. **Heuristic Shortcut:** If the `extract_icd_from_literal` function detects a valid ICD-10 or mapped ICD-9 code directly in the string, the model bypasses statistical inference and assigns the extracted category immediately.

2. **Statistical Inference:** If no explicit code is found, the normalized string is vectorized and passed to the trained `LinearSVC` for prediction.

3. **Knowledge Base Fallback:** For instances where the classifier yields no prediction, the system relies on a cosine similarity match against the official ICD-10 dataset, utilizing an independent word-level TF-IDF vectorizer (n-grams 1-2, 10,000 features).

3 Experiments and Results

3.1 Baseline Comparisons on Local 80/20 Split

Prior to training on the full dataset, a series of controlled experiments were conducted on an 80/20 stratified split (190,702 training samples, 2,740 validation samples) to evaluate different vectorization strategies, classifiers, and dimensionality reduction techniques.

Vectorization and classifier selection. Two different vectorization approaches were studied. In both of them, word-level and character-level TF-IDF features were combined using the `FeatureUnion` transformer. In the first approach, word n-grams ranging from unigrams to bigrams and character n-grams ranging from unigrams to 4-grams (the maximum number of features equals 10,000 for word-level and 20,000 for character-level) were utilized; while in the second one, the number of n-grams was increased to trigrams for word-level and 6-grams for character-level (20,000 and 60,000 features, respectively). These two models were tested using the `LinearSVC` classifier as a base model.

LinearSVC vs. Logistic Regression. Performance improved when using LinearSVC rather than Logistic Regression with vectorization conditions remaining the same. Where Logistic Regression resulted in accuracies of **55.26%** and **55.11%** on the two sets of vectorized instances, the hinge loss approach employed by LinearSVC showed to be much more suitable for the task at hand.

Effect of dimensionality reduction via SVD. Using `TruncatedSVD` (components = 300) and then applying L2 normalization before using `LinearSVC` model resulted in a steep decrease in validation accuracy to **44.23%**. Although SVD helps save memory and time while building models, it fails to consider the subtle differences in the language that are necessary to distinguish between similar ICD-10-ES chapters.

Hierarchical classification. In addition, the experiment was performed for a hierarchical classification approach involving two levels where, at Level 1, samples were divided into *disease* and *procedure*, and then specific classifiers were used to classify subcategories at Level 2. In this case, the accuracy of the binary classifier at Level 1 was **84.67%**, which means that the disease/procedure classification can be successfully inferred from lexical information on the surface level. At the same time, the accuracy of the validation was **57.23%**, which is the same as for the flat baseline. Overall, this approach did not improve the model, as observed upon submission to the Kaggle leaderboard.

Per-class analysis. Across all configurations, consistent patterns emerged in per-class performance. Categories with distinctive and frequent terminology, such as B (infectious diseases), Z (factors influencing health status), M (musculoskeletal), and I (circulatory), achieved F1 scores above 0.70. In contrast, numeric procedure categories (e.g., *6*, *7*, *2*) and rare classes (e.g., W, A) consistently yielded F1 scores below 0.25, reflecting both their lexical ambiguity and low support in the validation set.

The most frequent misclassifications reveal systematic and linguistically interpretable patterns. The dominant confusion, $6 \to O$ (46 cases), reflects the lexical overlap between procedure codes for the digestive system (chapter 6) and obstetric conditions (chapter O), both sharing abdominal and gastrointestinal terminology. The mutual confusion of V and Z (42 and 30 instances respectively) is especially interesting: as neither chapter refers to primary diseases but rather only supplementary factors of illness, their medical terminology differs hardly at all in lexical terms.

Table 2 summarizes the experiments conducted and presents the validation accuracy of each variation of the model.

Table 2: Validation accuracy on the local 80/20 split for different model configurations.

Configuration	**Classifier**	**SVD**	**Val. Acc.**
Word (1,2) + Char (2,4)	`LinearSVC`	No	**0.5752**
Hierarchical (disease/procedure)	`LinearSVC`	No	0.5723
Word (1,3) + Char (2,6)	`LinearSVC`	No	0.5690
Word (1,3) + Char (2,5)	`LogisticRegression`	No	0.5526
Word (1,2) + Char (2,4)	`LogisticRegression`	No	0.5511
Word (1,2) + Char (2,4)	`LinearSVC`	Yes	0.4423

3.2 Final Model and Leaderboard Results

These preliminary experiments established that a sparse linear model with hinge loss natively outperformed standard Logistic Regression, and that SVD-based dimensionality reduction was detrimental to performance. The final proposed architecture was trained on the fully concatenated dataset (201,809 rows). By integrating the CodiEsp dataset and the official ICD mappings, the model effectively learned the boundaries of all 36 target classes. Upon applying the prioritized prediction pipeline to the 6,667 hidden leaderboard samples, the system successfully distributed predictions across the highly imbalanced target space (heavily favoring categories 'O', 'Z', and '0').

4 Discussion and Limitations

The integration of a unified `FeatureUnion` containing up to 60,000 combined word and character sub-word features proved highly resilient to clinician typos and morphologic variance. By removing the strict hierarchical routing present in previous iterations of our approach, the flat `LinearSVC` avoided compounding routing errors, leveraging the full feature space directly against all 36 classes simultaneously.

However, resolving ambiguities across certain semantic boundaries remains a limitation of term-frequency models. For example, chapters such as *'V'* (External causes of morbidity) and *'Z'* (Factors influencing health status) share heavily overlapping vocabulary profiles regarding preventive encounters. Sparse linear models occasionally struggle to disambiguate these edge cases purely through word counts, suggesting that future improvements could incorporate dense semantic embeddings (e.g., clinical BERT variants) for deeper contextual understanding.

5 Conclusions

We successfully implemented a scalable NLP pipeline capable of categorizing highly unstructured, bilingual clinical notes into ICD-10 classifications. By prioritizing rule-based domain knowledge (regex matching and abbreviations replacement) alongside a robust flat LinearSVC trained on a hybrid word/character TF-IDF feature space, the system mitigates the inherent noise of hospital data. This architecture strikes an optimal balance between computational efficiency and classification accuracy in the medical informatics domain.

References

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